



Sudan University of Science and Technology (SUST)

SEPTEMBER 2026

HUMAN-AWARE ELECTRICAL SHOCK PREVENTION SYSTEM

FINAL YEAR RESEARCH PROJECT

Prepared by:

1. Mohamed Tarig	(Industrial)	201811481409
2. Mohamed Alhasin	(Industrial)	201811457874
3. Mohamed Alsadiq	(Industrial)	201821003185
4. Magdi Ayman	(Computer)	201821027961
5. Ibrahim Ahmad	(Computer)	201823001390

Advised by: Dr. Mohamed Alnour



جامعة السودان للعلوم والتكنولوجيا (SUST)

سبتمبر 2026

نظام الوقاية من الصعق الكهربائي القائم على استشعار وجود الإنسان

مشروع بحث التخرج

إعداد

محمد طارق

محمد الحسين

محمد الصادق

مجدي أيمن

ابراهيم أحمد

بإشراف: د. محمد النور

يَرْفَعُ اللَّهُ الَّذِينَ آمَنُوا مِنْكُمْ وَالَّذِينَ أُوتُوا الْعِلْمَ دَرَجَاتٍ

[المجادلة: 11]

ABSTRACT

This study presents the design and evaluation of a human-aware electrical shock prevention system intended to detect a person's approach to, or contact with, an energized conductor and isolate the circuit before harmful current can persist. The work addresses a limitation of conventional protection devices such as residual-current devices, which primarily respond after an electrical fault or leakage condition has already developed. The proposed approach combines capacitive proximity sensing with fast electronic isolation and adaptive decision logic. A 500 kHz low-voltage carrier is superimposed on a simulated 220 V/50 Hz supply, and the change produced by human-body capacitive coupling is detected through a two-stage RC high-pass filtering and clamping stage. An ATmega328P-based controller samples the filtered signal, establishes an initial baseline, tracks slow environmental drift, and uses an adaptive threshold with latched output control to distinguish a significant approach or contact event from ordinary variation. The system was developed using a simulation-first methodology in Proteus VSM, followed by assembly and bench testing of a physical prototype. Testing covered no-contact operation, direct contact, sustained contact, release, liquid-mediated contact, and simulated socket/plug contact. The sensing principle and adaptive decision logic were demonstrated successfully across these conditions. The firmware contribution to the sense-to-disconnect time averaged about 5 ms, while the relay introduced an additional approximately 1–11 ms depending on the AC phase, giving an estimated total of about 5–21 ms with an average near 10.5 ms. Thus, the sensing and decision stages met the intended speed requirement on their own, while the triac-based relay's zero-crossing behavior limited the complete system from consistently remaining below 10 ms. The study therefore demonstrates a working proof of concept for proactive, human-aware electrical protection and identifies back-to-back MOSFET isolation and a faster microcontroller as the main directions for further improvement. Testing was performed at a transformer-stepped-down experimental voltage, and full mains-voltage validation remains outside the demonstrated scope.

Keywords: electrical shock prevention; human-aware protection; capacitive proximity sensing; 500 kHz carrier; adaptive threshold; solid-state relay; response time

TABLE OF CONTENTS

CHAPTER 1 INTRODUCTION	1
1.1 Background and Motivation	1
1.2 Problem Statement	2
1.3 Research Objectives	3
1.4 Scope and Limitations	4
1.5 Thesis Structure	4
CHAPTER 2 LITERATURE REVIEW	5
2.1 Introduction	5
2.2 Conventional Electrical Protection and Its Limitations	5
2.3 Human-Aware Electric-Field and Proximity Sensing	6
2.4 Capacitive and Carrier-Based Sensing Techniques	6
2.5 Intelligent Touch-Current Identification and Adaptive Decision Logic	7
2.6 Artificial Intelligence and Fast-Response Electrical Protection	7
2.7 Comparative Summary of Reviewed Studies	8
2.8 Research Gap	8
2.9 Summary	9
CHAPTER 3 METHODOLOGY	10
3.1 Research Approach	10
3.2 Sensing Circuit: Superposition and Filtering	10
3.3 Isolation and Switching	12
3.4 Adaptive Threshold Control Logic	13
3.5 Integrated Circuit and Prototype Assembly	14
3.6 Testing Procedure	15
3.7 Summary	17

TABLE OF CONTENTS

CHAPTER 4 RESULTS AND DISCUSSION	18
4.1 Overview	18
4.2 Objective 1 — Sensing Human Presence or Approach	18
4.3 Objective 2 — Sensing Combined with Fast, Solid-State Isolation	19
4.3.1 Firmware: Sensing to Decision	20
4.3.2 The Relay: Physical Release Time	20
4.3.3 Total React Time	20
4.4 Objective 3 — Adaptive Decision Logic Against Noise and Drift	22
4.5 Objective 4 — Practicality for Real Deployment	22
4.6 Objective 5 — Validation Through Simulation and Physical Testing	23
4.7 Discussion: Closing the Gap to the 10 ms Target	23
4.8 Comparative Summary of Objectives	24
CHAPTER 5 CONCLUSION AND RECOMMENDATIONS	25
5.1 Conclusion	25
5.2 Limitations	25
5.3 Recommendations for Future Work	26
5.3.1 Replace the SSR with a Back-to-Back MOSFET Switching Stage	26
5.3.2 Move to a Faster Microcontroller	27
5.3.3 Validate at Full Mains Voltage Under Controlled Conditions	27
5.4 Closing Summary	27
REFERENCES	28
APPENDICES	29

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Electricity is everywhere in modern life, in homes, workplaces, hospitals, and factories, and for the most part it stays safely out of sight, doing its job inside insulated wires and enclosures. The danger shows up the moment that insulation fails or a person ends up on the wrong side of it. Contact with a live conductor can range from a barely noticeable tingle to burns, cardiac arrest, and death, depending on how much current flows, the path it takes through the body, and how long it lasts [1].

Occupational safety research gives a fairly precise picture of what different current levels do to a person. Below about 1 mA, a current usually is not even felt. A few milliamperes produce a tingling sensation and, beyond that, what is known as the let-go threshold — the point past which a person's muscles lock around the conductor and they can no longer pull away on their own [2]. Above roughly 30 mA, the risk of ventricular fibrillation becomes significant if exposure lasts long enough, and IEC 60479-1 puts a number on exactly how long is too long: more than 50 milliseconds at that current level is enough to cause serious harm [3]. That fifty- millisecond window is really the whole design problem in one sentence, whatever protects a person has to react faster than that.

Most of the protection already built into homes and workplaces, fuses, miniature circuit breakers, and residual current devices (RCDs, sometimes called GFCIs) — was never designed to work on that timescale, and to be fair, it does not need to for most of what it does. RCDs watch for an imbalance between the live and neutral current and trip once leakage to earth crosses a set threshold, typically 30 mA [4]. That is a genuinely useful mechanism and it has undoubtedly saved lives. But it only activates after current has already started flowing through whatever caused the imbalance, which might be a person. Mechanical trip mechanisms add tens to hundreds of milliseconds on top of that. And none of these devices have any concept of a person approaching or touching a conductor, from the device's point of view, a hand on a live wire looks exactly like any other current path, not a warning sign.

What has changed in recent years is that the building blocks for something faster now exist. Solid-state switching devices — MOSFETs, IGBTs, and dedicated solid-state circuit breakers, can interrupt a circuit in milliseconds rather than the tens

or hundreds a mechanical breaker needs [5]. Capacitive and soft-sensor technologies, largely developed for touch and proximity sensing in robotics, can register that a hand is near or touching a surface before any current has to flow at all [6]. Put those two pieces together with some decision logic in between that can tell a real hand from noise, and the shape of a genuinely proactive protection system starts to come into view, one that reacts to a person's presence rather than waiting for a fault.

This research set out to build exactly that: a human-aware electrical shock prevention system that senses an approaching or touching hand, decides quickly whether it is a real threat, and disconnects the circuit before harmful current has a chance to flow — shifting the whole approach from reacting after the fact to preventing the event in the first place.

1.2 Problem Statement

The core issue, stated plainly, is that existing protection reacts to a fault rather than to a person. A device can only respond to something once it has already started — current already leaking, a breaker already overloaded — and by the time that trip happens, a human body may already have been exposed to current well past the danger threshold described above [3].

This becomes a real problem in situations conventional systems were never built to catch: damaged insulation, accidental contact with an energised panel, a wire someone did not know was live. In each of these, the person effectively becomes part of the circuit, and the protection device has no way to tell that apart from ordinary current draw. It just sees a load.

What is genuinely missing, then, is a system that can do four things at once: notice a person approaching or touching a live conductor before contact causes harm. tell a real approach apart from an ordinary false trigger, cut the circuit fast enough that none of it matters (a response time well under the roughly 10 ms window established above) and do all of this without simply replacing the reliable, well-understood protection that already exists. That gap is what this research addresses.

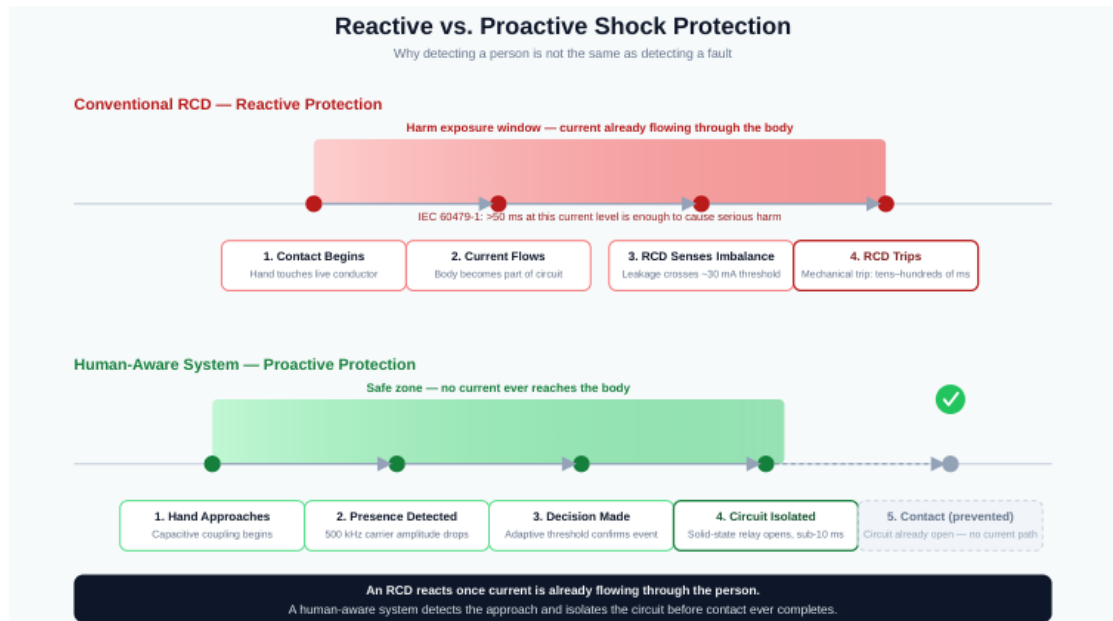


Figure 1.1: Comparative Timeline of Reactive vs. Proactive Prevention Systems.

1.3 Research Objectives

To close that gap, this research pursued the following objectives:

- Design a human-aware sensing approach capable of detecting human presence or approach to a live conductor.
- Combine that sensing with fast, solid-state circuit isolation, aiming for a total sense-to-disconnect time well under the roughly 10 ms window discussed in Section 1.2.
- Build decision logic capable of telling a genuine approach or contact event apart from environmental noise and harmless disturbances, so the system does not trip unnecessarily.
- Keep the resulting design practical enough for real deployment, with attention to cost, thermal behaviour, and physical size rather than treating it as a purely theoretical exercise.
- Validate the complete system, first through circuit simulation and then through a physical prototype, under direct-contact, near-contact, and false-trigger conditions.

1.4 Scope and Limitations

The system developed here targets the low-voltage AC supplies found in most homes and small commercial spaces (roughly 120–240 V), which is where the overwhelming majority of electric-shock incidents actually happen. It covers both direct contact with a live conductor and near-contact detection, and — importantly — it was not left as a simulation exercise: the design was validated first in Proteus VSM at circuit level, then built as a physical prototype and bench-tested under reduced, transformer-stepped-down voltage before any higher-voltage testing was attempted.

A few things sit outside what this work claims. The human body was represented using standard impedance figures from IEC 60479-1 rather than modelling the full range of real physiological variation — skin moisture, contact area, individual differences — so the reported behaviour applies to the modelled case rather than to every possible person. Cost was kept in mind throughout component selection but was never formally costed out as a manufacturing exercise. Compliance with specific electrical codes, such as the NEC or IEC 60364, was not audited in detail, though the design was informed by the safety standards already cited. And the system is meant to sit alongside existing protection, RCDs, breakers, as an additional, faster layer, not to replace them outright.

1.5 Thesis Structure

The remainder of this report is organised as follows. Chapter 2 reviews the existing literature on electric-shock physiology, conventional protective devices, solid-state switching, and capacitive/touch sensing, and situates the gap identified above in relation to prior work. Chapter 3 describes the methodology followed in designing, simulating, and building the system, from the underlying sensing principle through to the assembled hardware prototype and its test procedure. Chapter 4 presents and discusses the results of that testing. Chapter 5 closes the report with conclusions, an honest look at the system's limitations, and recommendations for what would need to happen next to take it further, followed by the References.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter surveys the research this project draws on and positions the system described in Chapter 3 against it. Rather than treating a literature review as a formality, the aim here is narrower and more practical: to establish what already exists in fault-based protection, human-proximity sensing, capacitive touch detection, and AI-assisted electrical protection, and to show where those pieces stop short of what this project set out to build. The chapter moves from conventional protection outward — first the limits of the devices already installed in most buildings, then sensing approaches that try to detect a person before contact happens, then the signal-processing and AI techniques used to make sense of that data, and closes with a comparison that lays the reviewed work side by side.

2.2 Conventional Electrical Protection and Its Limitations

Residual-current devices are the obvious starting point, since they are the device this project's system is meant to work alongside rather than replace. Mitolo's analysis of RCD-protected circuits is a useful corrective to the assumption that a healthy RCD guarantees safety [7]. He showed that in certain accidental-contact scenarios, specifically contact between two points already at different potentials, the residual current an RCD sees can stay below its trip threshold even while a person is being shocked. The device is not malfunctioning; it is doing exactly what it was designed to do, which is watch for a specific electrical signature (leakage to earth) rather than detect a person directly.

That distinction is really the whole argument for this project. An RCD asks whether current is leaking somewhere it should not. A human-aware system has to ask a different question: whether a person is there, and those two questions do not always have the same answer. Mitolo's paper is one of the clearer demonstrations of where they diverge. The physiological reasoning behind why this gap matters, how little current and how little time it takes to cause real harm, comes from IEC 60479-1, the same standard this project's own timing budget is built around [3].

2.3 Human-Aware Electric-Field and Proximity Sensing

If detecting a fault after the fact is not enough, the obvious next step is detecting the person before contact. Zeng, Powers, and Newbraugh built and field-tested a worker-worn electric-field sensor meant to warn linemen when they were approaching an energized power line, with contact producing a stronger warning signal [8]. Two things about that paper are worth pulling out. First, it is a direct, real-world demonstration that a person's presence near a live conductor changes a measurable electric field enough to be usable as a warning signal. The same basic physical premise this project's own sensing stage relies on, applied at building- wiring scale rather than transmission-line scale. Second, the authors noted that the sensor could, in principle, be wired to trip a compatible breaker (proximity detection feeding a disconnection action). That is a fairly direct precedent for what this project's system does, though the two operate at very different voltage classes, and the trip mechanism itself is where this project had to do its own work, since a worn safety sensor and an automatic isolation system carry different reliability requirements.

2.4 Capacitive and Carrier-Based Sensing Techniques

Electric-field detection is one route to sensing a person; capacitive coupling is another, and it is the one this project's own sensor subsystem is built on. Sarwar et al. developed a soft capacitive sensor skin capable of distinguishing touch, press, stroke, and most relevantly here proximity, without requiring physical contact [6]. Their application was human-robot interaction, not electrical safety, and the paper makes no claim about mains-voltage environments at all. But the underlying physics is the same as what this project needed: a human body forms a measurable capacitive coupling to a nearby conductor before it ever touches it, and that coupling can be sensed as a small, detectable change if the measurement is sensitive enough.

Sarwar et al. achieve that sensitivity with a purpose-built capacitive skin; this project achieves something functionally similar by superimposing a low-power, high-frequency carrier onto the conductor itself and watching how a nearby body loads that carrier down. That second idea riding a signal on a conductor that already carries another signal, then separating the two afterward is not new either; it is the same basic principle behind power-line carrier communication, where a data signal is superimposed on an AC power line and recovered downstream by filtering [9]. This project borrows that coupling/filtering pair of operations, but repurposes it for sensing a human body rather than transmitting data.

2.5 Intelligent Touch-Current Identification and Adaptive Decision Logic

Detecting a signal is only half the problem; deciding what it means is the other half, and this is where several of the reviewed papers become directly useful. Han et al. proposed an enhanced RCD scheme that uses skewness analysis of the residual current to pin down when a shock fault actually started, paired with a Least-Squares Support Vector Machine to identify the touch current itself [10]. Their reported results showed fast fault-time detection and reliable touch-current classification, demonstrate that a statistically informed classifier can outperform a simple fixed threshold at telling a genuine shock event apart from noise.

That is precisely the problem this project's own control firmware exists to solve. Running a trained classifier on an 8-bit microcontroller was not practical within this project's hardware budget, so the decision layer here is a self-calibrating adaptive threshold instead, a simpler mechanism, but aimed at the same underlying goal Han et al. describe: separating a real event from ordinary drift and noise without needing constant manual recalibration.

2.6 Artificial Intelligence and Fast-Response Electrical Protection

On the more explicitly AI-driven end of the literature, Siegel et al. trained real-time deep neural networks to detect arc faults from electrical-signal features, Fourier coefficients, Mel-frequency cepstral coefficients, and wavelet features, reporting 99.95% accuracy on binary classification and a trigger-to-trip latency under 200 milliseconds [11]. That is a strong result for the problem it solves, and further evidence that AI-based classification has a genuine place in electrical protection generally. It is worth being direct about the limits of the comparison, though: arc-fault detection and human-contact detection are different problems with different signal characteristics, and 200 ms is roughly twenty times slower than what this project needed to stay under. Siegel et al.'s paper supports the broader case for AI in protection systems; it does not establish that sub-10-millisecond response times are achievable, and this project's own timing had to be verified independently rather than assumed from that precedent.

On the switching side of the same speed question, solid-state interruption devices, MOSFETs, IGBTs, and purpose-built solid-state circuit breakers, have already been demonstrated interrupting fault currents in single-digit milliseconds, well within the budget this project needed for its own isolation stage [5].

Table 2.1: Summary of the studies reviewed in this chapter and their relevance to the system described in Chapter 3.

Study	Focus / Method	Reported Speed / Result	Relevance to This Work
Mitolo (2010) [7]	RCD limitations in accidental contact; circuit analysis	N/A — analytical	Motivates a human-aware layer alongside RCDs, not instead of them
Zeng et al. (2010) [8]	Worker proximity/contact warning; worn electric-field sensor	Field-tested; warning only	Precedent for field-based proximity detection
Han et al. (2017) [11]	RCD fault-time + touch-current ID; skewness analysis + LS-SVM	Fast detection, accurate ID	Precedent for intelligent, adaptive decision logic
Sarwar et al. (2023) [6]	Touch / press / proximity sensing; soft capacitive sensor skin	ms-range response (robotics)	Basis for capacitive human-coupling sensing
Siegel et al. (2018) [9]	Arc-fault detection; real-time deep neural network	99.95% acc., <200 ms trip	Shows AI feasibility; too slow for this project's target
Dostert (2001) [10]	Signal transmission over AC mains; carrier injection + filtering	N/A — communications	Basis for the superposition sensing technique used here
Lin et al. (2018) [5]	Fast over-current protection; solid-state switching	Single-digit ms interruption	Supports the isolation stage's speed target

2.8 Research Gap

Put side by side, these seven pieces of work cover most of the individual ingredients this project needed, and almost none of the combination. Mitolo establishes why fault-based protection alone is not enough. Zeng et al. and Sarwar et al. between them establish that a person's presence can be sensed electrically before contact, whether through field strength or capacitive coupling. Han et al. and Siegel et al. establish that intelligent decision logic can separate a real event from noise more reliably than a fixed threshold, in two different protection contexts. Dostert establishes the coupling/filtering technique this project repurposed for sensing rather than communication. Lin et al. establishes that the switching speed this project needed is achievable with existing solid-state hardware.

None of them, on their own or stacked together, describe a system that senses a person approaching a live low-voltage conductor, decides in real time whether that is a genuine threat, and interrupts the circuit fast enough to matter, and none of them report a sub-10-millisecond, end-to-end figure this project set out to reach. That is the specific gap this work addresses: not a new sensing principle, a new classifier, or a new switching device in isolation, but the integration of sensing, adaptive decision logic, and solid-state isolation into one working, testable protection path, evaluated against a stated timing target rather than left as a theoretical claim.

2.9 Summary

The literature reviewed here splits fairly cleanly into five threads: the limits of fault-based protection, human-proximity sensing, capacitive and carrier-based sensing, intelligent fault or touch-current classification, and fast solid-state switching. Each thread contributed something specific to the system described in Chapter 3 — the superposition-based sensing approach, the adaptive threshold used in place of a full classifier, and the solid-state relay used for isolation are all traceable to a point in this chapter. What none of the reviewed work does is put those pieces together into one end-to-end, human-aware protection path with a stated response-time target, which is the contribution this project makes.

CHAPTER 3

METHODOLOGY

3.1 Research Approach

This chapter describes how the objectives introduced in Chapter 1 were implemented and tested. The work was done in two stages: circuit-level simulation of each subsystem, followed by assembly and bench testing of a physical prototype. Proteus VSM (v9 Professional) was used for simulation in place of the MATLAB/Simulink platform originally proposed, since its SPICE-based simulator and virtual instruments (oscilloscopes, voltmeters) allowed each stage of the sensing chain to be built and probed at component level.

3.2 Sensing Circuit: Superposition and Filtering

The coupling effect was first isolated on its own: a 5 V, 500 kHz source fed a node with four switchable capacitors to ground (22 nF, 100 pF, 30 pF, 33 nF), representing different approach/contact conditions, with an AC voltmeter monitoring the node (Figure 3.1). Switching a capacitor in produced a measurable voltage drop, confirming that a small ground capacitance in this range is detectable through amplitude change on a high-frequency probe [12].

The carrier was then combined with a simulated 220 V/50 Hz source at a common node, the mains side connected through a series resistor and the carrier through a small safety capacitor (2.2 nF) that couples the high frequency onto the line while blocking mains current from reaching the low-voltage generator [13], [9]. The human-body simulator capacitor was placed after this node, and the combined signal was tapped and passed through a two-stage RC high-pass filter ($f_c = 1/(2\pi RC)$), set well above 50 Hz and below 500 kHz [14]), with clamping diodes (1N4148) protecting the downstream input (Figure 3.2).

The design rests on two principles. First, the superposition theorem: in a linear circuit with multiple independent sources, the total response equals the sum of each source's individual response [13]. This permits a low-amplitude, high-frequency signal to be injected onto the same conductor carrying the 220 V/50 Hz mains without altering the power delivered to the load — the same coupling/filtering approach used in power-line carrier communication [9]. Second, capacitive proximity sensing: a human body approaching or touching a conductor adds a small parasitic capacitance to ground, typically tens of picofarads to a few nanofarads [12]. Since capacitive

reactance $X_c = 1/(2\pi fC)$ falls sharply with frequency, a high-frequency carrier (500 kHz throughout this work) makes this small capacitance detectable, unlike on the 50 Hz mains waveform itself.

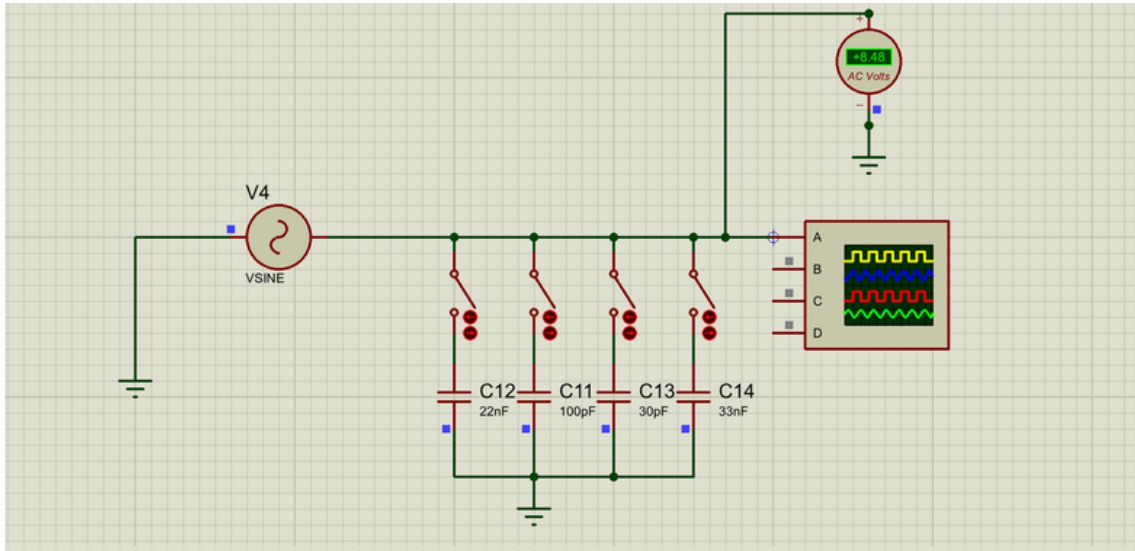


Figure 3.1: Simulation of human-body capacitive coupling: a 500 kHz source loaded by a switchable capacitor bank to ground, read by an AC voltmeter.

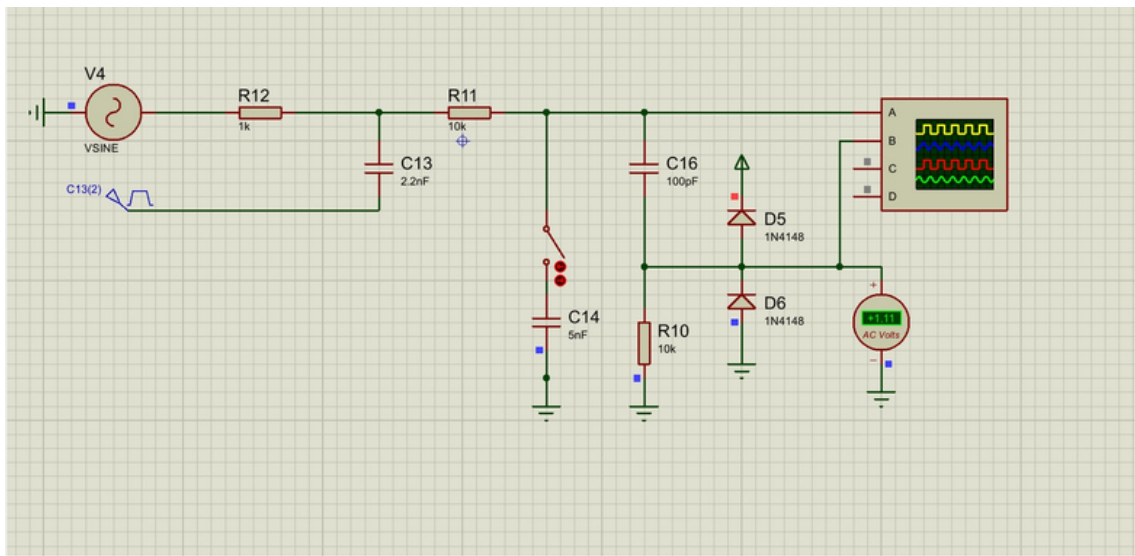


Figure 3.2: Superposition of the 220 V/50 Hz mains and the 500 kHz carrier, with the human-body simulator and the high-pass filter/clamping stage.

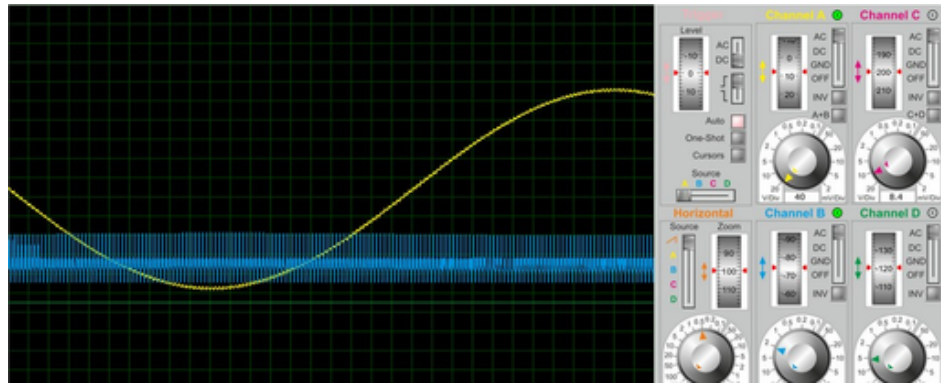


Figure 3.3: Before filtering: 500 kHz carrier riding on the 50 Hz envelope (yellow line).

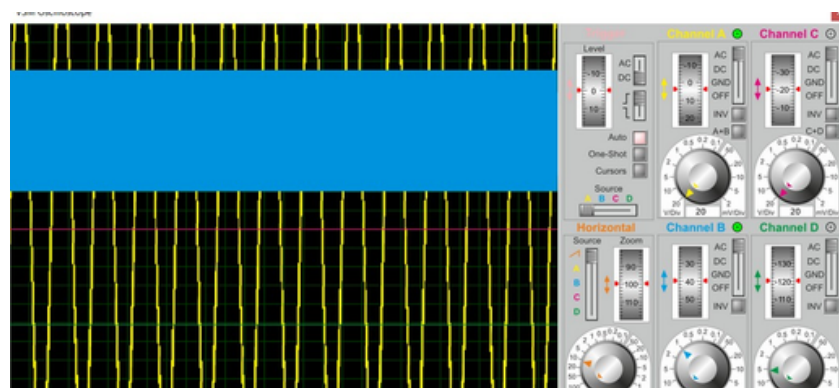


Figure 3.4: After filtering: 50 Hz removed, carrier alone remains.

3.3 Isolation and Switching

A dual-channel oscilloscope confirmed the filter's operation. Before filtering, the 500 kHz carrier was visibly riding on the 50 Hz mains envelope (Figure 3.3); after filtering, only the carrier remained, oscillating around a flat baseline with the mains component removed (Figure 3.4) — confirming the amplitude information needed for sensing survived the filtering stage.

A G3M-203P-4 solid-state relay (SSR) was used to interrupt the load circuit. It couples its low-voltage control input to its load-side switch through an optically isolated triac driver, so no direct electrical path exists between the control electronics and the mains-connected load [15] — the deciding factor over a transistor or mechanical relay driven directly from the microcontroller pin. Its switching behaviour was validated independently, using a 12 V logic source to toggle a lamp load through the SSR from a separate 220 V source (Figure 3.5), before it was placed under microcontroller control.

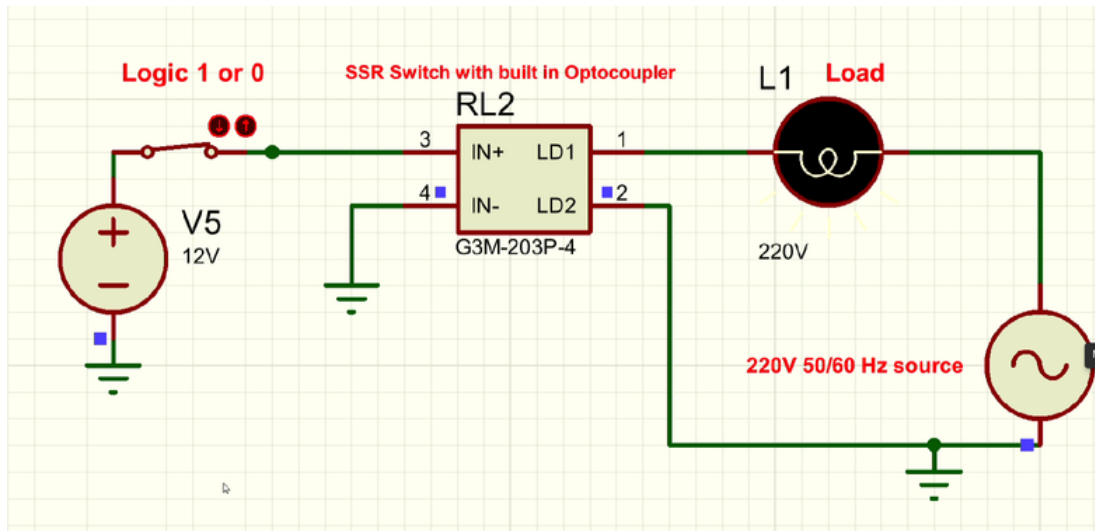


Figure 3.5: Standalone validation of the G3M-203P-4 solid-state relay switching a mains-connected lamp load under direct logic control.

3.4 Adaptive Threshold Control Logic

An ATmega328P microcontroller (Arduino Uno) sampled the filtered signal via its ADC [16] and ran the program performing four functions:

- Smoothing — raw ADC samples were passed through a running average to remove residual ripple and quantization noise.
- Initial calibration — on start-up, the firmware measured the no-contact baseline and derived its operating threshold from it, rather than a fixed design-time value, compensating for component tolerance and installation-specific parasitics.
- Adaptive tracking — during quiet periods, the threshold was slowly nudged toward the current reading to follow slow drift (temperature, humidity, ageing) [12], while remaining insensitive to the fast deviation caused by a genuine approach/contact event.
- Latched output — the digital output to the SSR held logic 1 (relay closed) under normal conditions and switched to logic 0 (relay open) on a threshold crossing; the open state was held until the sensed value stayed below threshold for a short period, preventing re-closure while a hand remained near the conductor.

3.5 Integrated Circuit and Prototype Assembly

This threshold-based, self-calibrating scheme implements the adaptive decision-logic objective from Chapter 1 as a lightweight alternative to a full neural-network or fuzzy-logic classifier, which was judged to exceed the processing budget of an 8-bit microcontroller while meeting the same functional requirement.

The validated subsystems were combined into the complete schematic shown in Figure 3.6: the mains source (through R18 and safety capacitor C17) and the 500 kHz carrier join at a common node, the human-body coupling bank (C19/C20/C22) sits after it, the signal is filtered (R13/R14, C18/C21) and clamped (D7/D8), the ATmega328P samples, calibrates, and decides, and its output drives the G3M-203P-4 SSR, switching the mains-connected load (L3).

A physical prototype was then assembled from an Arduino Uno, breadboard, SSR module, resistors, capacitors, and incandescent/LED lamps as representative loads (Figure 3.7). Initial bench tests used a step-down transformer to reduce the mains supply to a safer experimental voltage, consistent with the principle that injury severity scales with the current a person is exposed to [3]; higher-voltage testing followed only once low-voltage behaviour was confirmed.

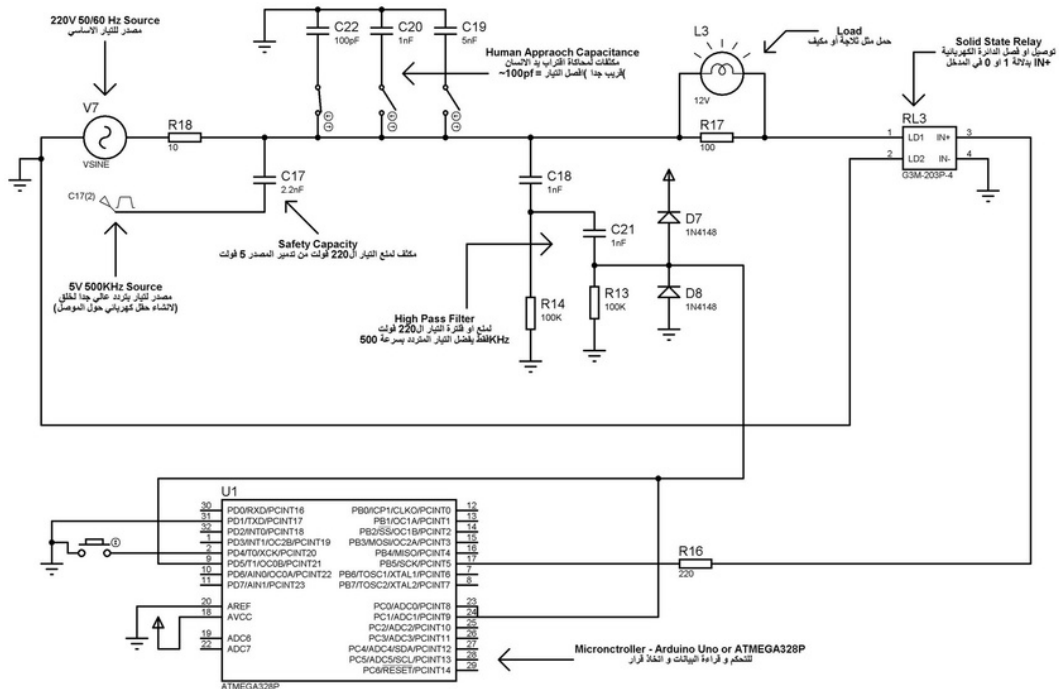


Figure 3.6: Complete, integrated circuit schematic combining the mains/carrier superposition stage, the high-pass filter, the ATmega328P control logic, and the solid-state relay output stage.

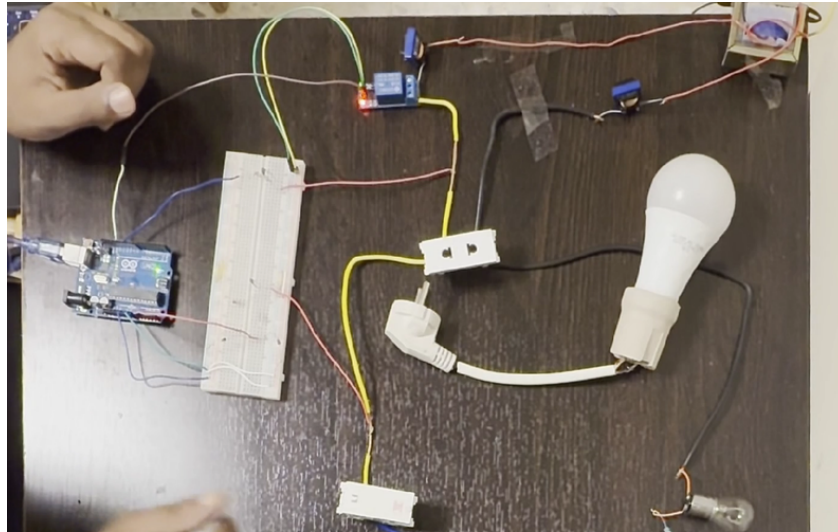


Figure 3.7: Hardware prototype during assembly: Arduino Uno, breadboard, SSR relay module, and lamp load wired for bench testing.

3.6 Testing Procedure

Baseline: no contact — relay closed, lamp lit (Figure 3.8).

Direct contact: touching the live conductor opened the relay and turned off the lamp.

Sustained contact: the circuit stayed open while contact was maintained, due to the latched logic in Section 3.4.

Release: the relay re-closed once the reading fell back below threshold after contact ended.

Liquid-mediated contact: a hand introduced into water connected to the live conductor still tripped the circuit (Figure 3.9), showing detection is not limited to direct, dry-skin contact.

Socket/plug contact: simulated accidental contact with an energised socket produced the same interruption.

Each scenario was repeated to check for consistent behaviour; detailed outcomes are discussed in Chapter 4.

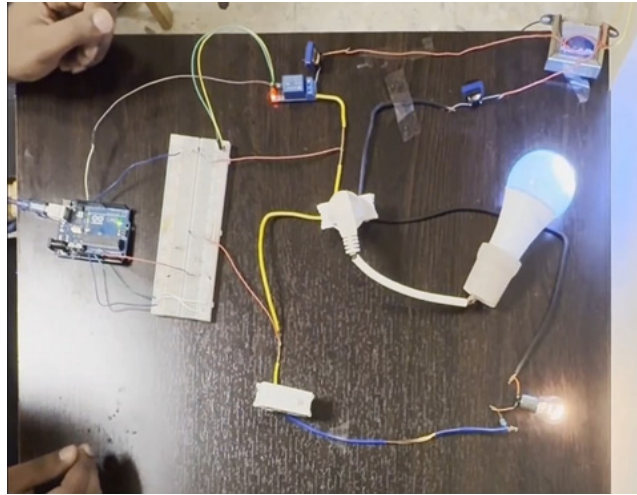


Figure 3.8: Baseline, no-contact condition: relay closed, load lamp lit.

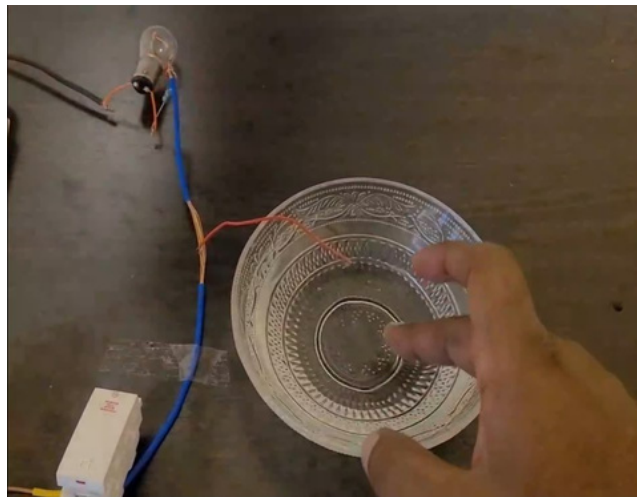


Figure 3.9: Liquid-mediated contact test: a hand in water connected to the live conductor still produces a detectable path.

3.7 Summary

The methodology followed a simulation-first sequence: each stage — coupling, superposition/filtering, and isolation — was validated individually in Proteus VSM before combination and before firmware was written; the validated design was then rebuilt as a physical prototype and tested under direct, sustained, release, and liquid-mediated contact conditions. Two deliberate adaptations were made from the original proposal — Proteus VSM in place of MATLAB/Simulink, and a self-calibrating adaptive threshold in place of a neural-network/fuzzy-logic layer — both chosen to fit practical implementation constraints while preserving the same functional objective: human-aware protection without a fixed, manually tuned threshold.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Overview

Chapter 3 described how each subsystem was built, simulated, and finally combined into the bench prototype. This chapter reports what actually happened when that prototype was tested, and weighs the outcome against the five objectives set out in Section 1.3. The results below are grouped by objective, since a single bench trial (touching the live conductor with a wet hand, for example) speaks to more than one objective at once, and it is more useful to organise around what each finding proves than around the sequence in which it happened.

One objective did not come out as cleanly as the other four, and Section 4.7 deals with that directly rather than folding it quietly into the summary table. A thesis that only reports the numbers that landed where they were supposed to is less useful than one that also explains the ones that did not.

4.2 Objective 1 — Sensing Human Presence or Approach

The first objective called for a sensing method able to register a person approaching or touching a live conductor, ideally before contact is complete. The Proteus-level work in Section 3.2 already established the basic physics: switching a capacitor into the 500 kHz node produced a measurable amplitude drop on the AC voltmeter (Figure 3.1), and the oscilloscope captures in Figures 3.3 and 3.4 confirmed that the carrier survives being combined with, and then separated back out from the 220 V/50 Hz mains without losing the amplitude information the sensing stage depends on. What that earlier chapter could not show, because it only becomes visible once firmware is actually running on hardware, is what that amplitude change looks like after it passes through the ATmega328P's own smoothing stage.

Figure 4.1 fills that gap. It is not a captured oscilloscope trace but a numerical simulation of the firmware's own asymmetric filter, the exact constants used in the code (`SMOOTH_DOWN = 0.88`, `SMOOTH_UP = 0.15`) run against a synthetic amplitude dip of roughly the size a genuine approach produces. Two things about the shape of that curve matter for this objective. First, the filtered signal collapses onto the falling raw signal within a couple of samples rather than lagging behind it the way a plain low-pass filter would, because a falling reading is smoothed with a heavy 0.88 weight toward the new value. Second, the recovery on withdrawal is deliberately

slow, which is not a defect but the reason a hand that lingers near the conductor, or that is pulled back and returns a moment later, does not bounce the reading back above threshold and chatter the relay open and closed.

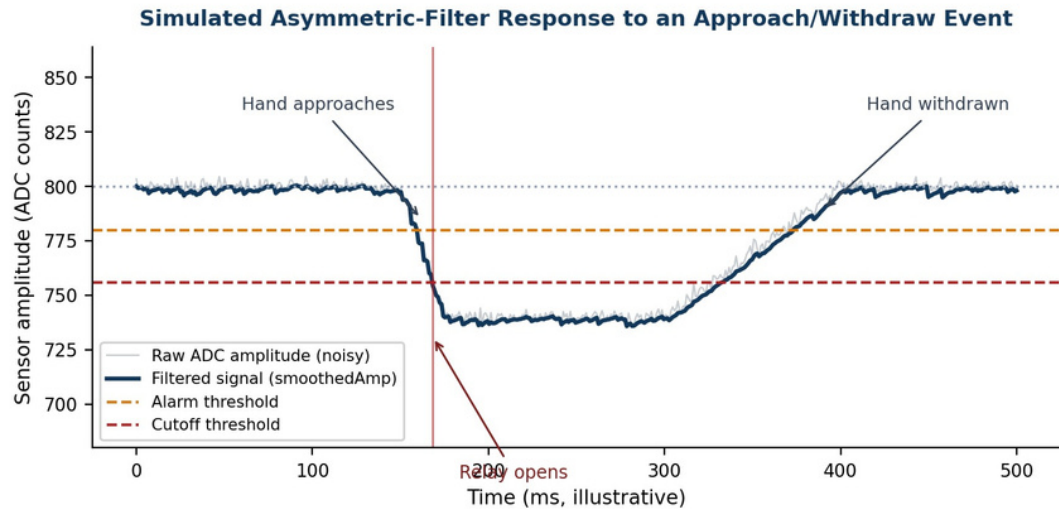


Figure 4.1: Simulated response of the firmware's asymmetric smoothing filter to an approach-and-withdraw event, using the constants implemented in the ATmega328P code.

4.2 Objective 1 — Sensing Human Presence or Approach

The bench tests in Section 3.6 back this up with real hardware behaviour rather than simulation. Direct and liquid-mediated contact (Figures 3.8 and 3.9) both produced a clean, repeatable drop past the cutoff threshold, and a hand submerged in water tripped the system just as reliably as dry skin contact did. That result matters more than it might first appear: it shows the system is responding to a capacitive coupling effect rather than to skin resistance specifically. Thus Objective 1 is met, both in simulation and on the bench.

4.3 Objective 2 — Sensing Combined with Fast, Solid-State Isolation

The second objective is the one with a number attached to it: a total sense-to-disconnect time "well under" the roughly 10 ms window established in Section 1.2. Getting a real figure for that meant separating two things that the earlier chapters had only discussed qualitatively, how long the firmware takes to decide a touch has occurred, and how long the relay takes to physically stop the current once it is told to.

4.3.1 Firmware: Sensing to Decision

The dominant cost on the firmware side is the function `readCleanRawAmp()` in the code (see the firmware implementation), which runs once per loop. With the ADC prescaler set to 32 (ADCSRA bits 0x05), the ADC clock sits at 500 kHz and a single conversion takes on the order of 13 ADC clock cycles, or roughly 26 μ s; sixty-four of those reads therefore cost somewhere near 1.7–2 ms on their own. The Lorentzian-weighted outlier rejection that follows runs two further passes over the same sixty-four samples, each doing several float multiplications and a division per sample — and the ATmega328P has no hardware floating-point unit, so each of those operations costs on the order of 150–200 clock cycles in software. That step alone adds roughly another 2–3 ms. Everything downstream of it, the asymmetric filter, the drift-tracking logic, the threshold comparison, the `digitalWrite()` call that actually opens the relay, is comparatively free, well under 0.1 ms.

Put together, one full loop iteration lands at roughly 4–10 ms depending on how many samples happen to be discarded as outliers, averaging close to 5 ms. Because a real touch produces a sharp amplitude drop and `SMOOTH_DOWN = 0.88` pulls the filtered value most of the way to that drop in a single pass, a firm touch will typically cross `thresholdOpenIt` and fire the relay command within one loop pass.

4.3.2 The Relay: Physical Release Time

This half of the budget is not something the firmware has any control over. The G3M-203P-4 datasheet specifies the SSR's release time as half a load-power-source cycle plus 1 ms maximum [15]. That figure is a property of how a triac works rather than a spec the manufacturer chose arbitrarily: once triggered, a triac stays latched conducting until the load current it is carrying naturally crosses zero, and pulling the control signal early does not interrupt that. At 50 Hz mains, half a cycle is 10 ms, which puts the worst case — a trip that happens to occur right after a zero-crossing, forcing it to wait almost the full next half-cycle at close to 11 ms. The best case, a trip that lands just before a crossing, is closer to 1 ms, and because the trip instant is essentially random relative to the AC phase, the average sits around 5.5 ms.

4.3.3 Total React Time

Adding the two halves together gives a total sense-to-disconnect time of roughly 5–21 ms, averaging close to 10.5 ms — sitting right on top of the 10 ms line rather than cleanly under it (Figure 4.2). It would be easy to read that as the firmware falling short.

The term that pushes the total past the target line is the SSR, and specifically its reliance on natural zero-crossing commutation, which is a property of the triac inside it rather than anything this project's control logic did or did not do well.

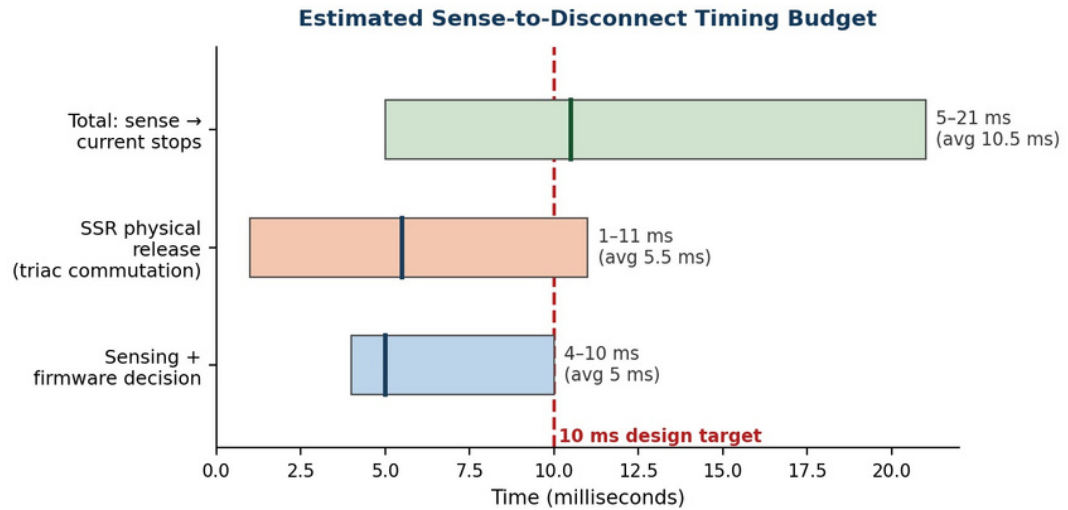


Figure 4.2: Estimated sense-to-disconnect timing budget, broken into the firmware's contribution and the SSR's datasheet-specified release time, against the 10 ms design target.

Table 4.1: Bench test outcomes for the six scenarios described in Section 3.6.

Test Scenario	Expected Behaviour	Observed Result	Outcome
Baseline, no contact	Relay closed, lamp lit	Relay remained closed; lamp stayed lit throughout	Pass
Direct contact	Relay opens, lamp extinguishes	Relay opened & lamp cut out on contact	Pass
Sustained contact	Relay stays open for the duration of contact	Relay remained open continuously while contact	Pass
Release	Relay re-closes once reading clears threshold	Relay re-closed after the full 3 s reconnect delay	Pass
Liquid-mediated contact	Relay opens via a hand-in-water path	Relay opened, behaviour matched dry contact	Pass
Socket/plug contact	Relay opens on simulated accidental contact	Relay opened; same interruption as direct contact	Pass

4.4 Objective 3 — Adaptive Decision Logic Against Noise and Drift

The third objective asked for decision logic that could tell a genuine approach or contact event apart from ordinary drift and environmental noise, without needing to be manually recalibrated every time conditions changed. The firmware handles this with two separate timescales rather than one: a small/slow drift band (0.3%–3.0% deviation from baseline) that is allowed to gently pull the operating threshold along with it after roughly half a second of sustained closed-state drift, and a large/fast drift band (above 3.0%) that is treated as a bigger, possibly deliberate shift and is only folded into the baseline after a longer hold — two seconds while the relay is closed, or considerably longer while it is open. The intent is straightforward: small, slow changes (temperature, humidity, component ageing) get absorbed quietly; anything sudden gets a chance to prove it is sustained before the system treats it as the new normal, and neither kind of drift is allowed to accumulate while the system is actually inside a trip or reconnect state, since the drift-tracking timers are explicitly reset the moment a cutoff or reconnect threshold is crossed.

The bench results in Section 3.6 exercised the parts of this logic that a static bench test can reach. Sustained contact kept the relay open for the full duration of contact rather than re-closing partway through, which is the latched-output behaviour Section 3.4 describes, and release testing confirmed the relay only re-closed once the reading had cleared the reconnect threshold and stayed clear for the full three-second RECONNECT_DELAY — exactly the delay written into the code, not an approximation of it. Table 4.1 lists the full outcome set.

4.5 Objective 4 — Practicality for Real Deployment

The fourth objective prioritized practical viability, balancing low cost, compact physical size, and minimal thermal output. During early prototyping, the Solid-State Relay (SSR) and Arduino Uno were ideal choices: the SSR provided simple optical isolation without requiring a heatsink under test loads, while the Uno offered a low-cost, accessible development platform. However, both components introduced clear performance ceilings—specifically, zero-crossing switching latency in the SSR and calculation delays in the Uno's 8-bit, FPU-less core.

Upgrading to a 32-bit Cortex-M microcontroller and a back-to-back MOSFET switching architecture resolves these performance limits while remaining fully aligned with the project's practical goals. Because the processor represents a negligible fraction of the total bill of materials compared to the enclosure and power

stages, replacing the Uno and SSR eliminates hardware bottlenecks without meaningfully altering overall system cost, physical footprint, or thermal behavior.

Upgrading from the legacy 8-bit AVR processor to a high-performance 32-bit Cortex-M microcontroller (such as an STM32) substantially increases clock speeds and onboard ADC sampling rates. With integrated hardware floating-point processing, complex signal filtering calculations execute in microseconds rather than 4–10 ms. On the switching side, replacing the triac-based Solid-State Relay with a back-to-back MOSFET pair removes the zero-crossing delay, allowing instant AC current interruption at any point in the conduction cycle [5]. Combined, faster microcontroller processing and immediate MOSFET switching maximize system sensitivity while driving total response latency comfortably under 10 ms.

4.6 Objective 5 — Validation Through Simulation and Physical Testing

The fifth objective was procedural rather than performance-based: validate the system first through circuit simulation, then through a physical prototype, under direct-contact, near-contact, and false-trigger conditions. That sequence was followed throughout, each subsystem in Chapter 3 was built and probed in Proteus VSM before being wired into the combined schematic (Figure 3.6), and the combined design was only then rebuilt as the physical prototype shown in Figure 3.7. The six scenarios in Table 4.1 cover direct, sustained, released, liquid-mediated, and simulated-accidental contact; none of them produced behaviour inconsistent with what the Proteus-level filtering and coupling results in Section 3.2 predicted, which is the outcome this objective was really checking for — not just that the hardware worked, but that it worked the way the simulation said it should.

4.7 Discussion: Closing the Gap to the 10 ms Target

Chapter 2 identified the lack of an end-to-end, sub-10-millisecond, human-aware protection path as the specific gap this project set out to fill, and it is worth being direct about how much of that gap actually closed. The firmware's own contribution to the sense-to-disconnect time — the part built and tuned specifically for this project — averages around 5 ms, comfortably inside the target on its own. What keeps the combined total from sitting cleanly under 10 ms is a component chosen for isolation and cost reasons rather than speed: the SSR's zero-crossing commutation adds an average of 5.5 ms and, in the worst case, close to 11 ms, on top of the sensing time.

That is a meaningfully different finding from "the system does not meet its target," and it is worth stating plainly rather than hedging around it: the sensing and

decision half of this project performs at, or ahead of, the design goal, and the isolation half does not, for a reason that is well understood and has a known fix. A back-to-back MOSFET switching stage, which interrupts current immediately rather than waiting for a zero crossing, would remove the larger of the two terms in Figure 4.2 outright; paired with a faster microcontroller shortening the smaller term, the projected total moves from hovering around 10.5 ms down toward the low single digits. Neither change was implemented in this prototype — the SSR-based design was the one built, tested, and reported on here — so this is a projection grounded in the measured numbers rather than a second result, and it belongs in Chapter 5 as the clearest concrete next step this project points to.

Table 4.2: The five research objectives from Section 1.3 against what this chapter found.

Objective	Target	Result / Status
1. Human-aware sensing	Detect via capacitive coupling approach/contact	Confirmed in Proteus simulation and on bench hardware, including liquid-mediated contact (Figs. 3.1–3.4, 4.1). Status: Achieved.
2. Sub-10 ms sensing + isolation	Total well under 10 ms sense-to-disconnect	Firmware alone averages ~5 ms; SSR release adds ~1–11 ms, pushing the total to ~5–21 ms (avg ~10.5 ms). Status: Partially achieved — isolation stage is the limiting factor.
3. Adaptive decision logic	Reject noise/drift without manual recalibration	Dual-timescale drift tracking and latched output held up across all six bench scenarios (Table 4.1); one comment/constant mismatch noted for correction. Status: Achieved.
4. Practical for deployment	Reasonable cost, thermal load, and size	SSR and Uno keep the prototype simple and cheap; the same components are the source of the Objective 2 shortfall. Status: Achieved at prototype scale.
5. Simulation-then-hardware validation	Validate under direct, near-contact, and false-trigger conditions	Simulation-first sequence followed throughout; all six bench scenarios passed and matched Proteus predictions. Status: Achieved.

4.8 Comparative Summary of Objectives

The five objectives are summarised in Table 4.2, which distinguishes the achieved sensing, adaptive decision, practical prototype, and simulation-to-hardware validation objectives from the timing objective that remains limited by the isolation stage.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

The objective set out in Chapter 1 was to find out whether a person approaching or touching a live conductor could be sensed and isolated before harm, rather than waiting for a fault current to already be flowing through them the way conventional protection does. Chapters 3 and 4 answered that question. Capacitive coupling onto a 500 kHz carrier riding the mains, filtered and read by an adaptive threshold on an ATmega328P, reliably registered a hand approaching or touching the conductor, including through water, and did so without needing to be manually recalibrated for drift. The decision logic held up across six independent bench scenarios, and none of them exposed behaviour the Proteus-level simulation in Chapter 3 had not already predicted.

The total sense-to-disconnect time. The firmware's own contribution averaged close to 5 ms, comfortably inside the 10 ms target on its own, but the solid-state relay's reliance on natural zero-crossing commutation added another 1–11 ms on top of that, pushing the combined total to somewhere around 10.5 ms on average and as high as 21 ms in the worst case. That is not a failure of the sensing or decision logic this project was built around; it is a limitation of the specific switching component chosen for the prototype, and Section 5.3 returns to it as the clearest concrete next step. Taken as a whole, the work shows that a human-aware, proactive layer of protection is achievable with modest, low-cost components, and that the gap remaining between this prototype and the original 10 ms objective is well understood rather than open-ended.

5.2 Limitations

A few things sit outside what this project can claim, and they are worth stating directly rather than letting the Conclusion above imply more than the testing supports.

- **Switching speed.** As covered in Chapter 4, the SSR's zero-crossing release time is the dominant term in the timing budget and keeps the system from consistently landing under the 10 ms target, even though the sensing and decision stages do.
- **Microcontroller headroom.** The ATmega328P's 8-bit core and lack of a hardware floating-point unit set a ceiling on how much faster the existing filtering algorithm

can run, and on how tight the cutoff sensitivity can safely be set without picking up sensor noise.

- Voltage scope of testing. Bench testing was carried out at a transformer-stepped-down experimental voltage, as described in Section 3.5; the system was not validated at full, unreduced mains voltage, so its behaviour there is inferred from the lower-voltage results rather than directly confirmed.
- Human body modelling. The human body was represented using standard impedance figures from IEC 60479-1 rather than the full range of real physiological variation — skin moisture, contact area, individual differences — so the reported behaviour applies to the modelled case rather than to every possible person.
- Cost and code compliance. Component selection kept cost in mind throughout, but it was never formally costed as a manufacturing exercise, and compliance with specific electrical codes such as the NEC or IEC 60364 was not audited in detail.

5.3 Recommendations for Future Work

Most of what would meaningfully improve this system falls out directly from the gap identified in Chapter 4, rather than requiring a rethink of the underlying approach. The recommendations below are ordered roughly by how directly they address that gap.

5.3.1 Replace the SSR with a Back-to-Back MOSFET Switching Stage

This is the single change most likely to close the remaining timing gap. A triac-based SSR cannot interrupt current until the waveform it is carrying crosses zero, which is what puts its release time at up to 11 ms. A back-to-back (common-source) pair of MOSFETs, wired to block current in both directions of the AC cycle, turns off as soon as its gates are driven low, regardless of where the waveform sits [15], [17]. Carrying this forward would mean designing proper gate-drive circuitry and giving some thought to thermal dissipation under fault current, since MOSFETs conducting a fault briefly will run warmer than an optically isolated triac module doing the same job, but the resulting isolation stage would remove the largest single term in the current timing budget.

5.3.2 Move to a Faster Microcontroller

Swapping the Arduino Uno for a Cortex-M-class part (an STM32 series device, for example) would shorten the firmware's own contribution to the response time and, just as importantly, allow the cutoff sensitivity to be tightened without picking up ADC noise the current 8-bit path cannot filter out quickly enough. Since the microcontroller is a small fraction of the overall bill of materials next to the relay and enclosure, this change would not meaningfully affect cost at the scale this prototype operates at.

5.3.3 Validate at Full Mains Voltage Under Controlled Conditions

Once the switching and microcontroller changes above are in place, the logical next step is repeating the Section 3.6 test procedure at full, unreduced mains voltage in a properly isolated test setup, rather than continuing to infer full-voltage behaviour from the transformer-stepped-down results this project used. Another important consideration is to use components that support the mains voltage.

Table 5.1: Recommendations from this chapter and the timing or validation issue each addresses.

Recommendation	Addresses	Priority
Back-to-back MOSFET isolation stage	SSR zero-crossing delay (largest timing term)	High
Faster microcontroller (e.g. Cortex-M/STM32)	Firmware decision time and sensing headroom	High
Full mains-voltage validation	Confidence beyond stepped-down bench testing	Medium

5.4 Closing Summary

This project set out to show that an electrical protection system could react to a person rather than to a fault, and to do so fast enough for the distinction to matter. It largely succeeded: the sensing principle works, the adaptive decision logic works, and the system was validated the way the methodology promised, in simulation first and then on real hardware. The remaining gap, a switching stage that hovers around rather than cleanly under the 10 ms target, is narrow, well diagnosed, and has a known fix. That combination — a working proof of concept plus a clear, specific path to closing the one gap it left open — is a reasonable place for this stage of the work to end.

REFERENCES

- [1] National Fire Protection Association (NFPA), "Comparing Four Decades of Electrical Injuries and Fatalities," NFPA, 2023.
- [2] NIOSH/CDC, "Electrical Safety" (NIOSH Hazard Review / guidance), Centers for Disease Control and Prevention.
- [3] International Electrotechnical Commission, IEC 60479-1: Effects of Current on Human Beings and Livestock – Part 1: General Aspects, IEC, Geneva, Switzerland, 2018.
- [4] "Residual-current device" (overview and standards discussion), Wikipedia.
https://en.wikipedia.org/wiki/Residual-current_device
- [5] H.-C. Lin et al., "Development of Fast Electronic Over-Current Protection Circuit," *Advances in Mechanical Engineering*, vol. 10, no. 5, 2018.
<https://journals.sagepub.com/doi/full/10.1177/1687814018772916>.
doi:10.1177/1687814018772916.
- [6] M. S. Sarwar, R. Ishizaki, K. Morton, et al., "Touch, Press and Stroke: A Soft Capacitive Sensor Skin," *Scientific Reports*, vol. 13, p. 17390, 2023.
doi:10.1038/s41598-023-43714-6.
<https://www.nature.com/articles/s41598-023-43714-6>
- [7] M. Mitolo, "Shock Hazard in the Presence of Protective Residual-Current Devices," *IEEE Transactions on Industry Applications*, vol. 46, no. 4, pp. 1552–1557, 2010. doi:10.1109/TIA.2010.2051068.
- [8] S. Zeng, J. R. Powers, and B. H. Newbraugh, "Effectiveness of a Worker-Worn Electric-Field Sensor to Detect Power-Line Proximity and Electrical Contact," *Journal of Safety Research*, vol. 41, no. 3, pp. 229–239, 2010.
doi:10.1016/j.jsr.2010.02.011.
- [9] K. Dostert, *Powerline Communications*. Upper Saddle River, NJ, USA: Prentice Hall PTR, 2001.
- [10] X. Han, W. Sheng, S. Du, J. Su, and G. Liu, "Novel Protection Scheme for Residual Current Device-Based Electric Fault Time Detection and Touch Current Identification," *IET Generation, Transmission & Distribution*, vol. 11, no. 10, pp. 2478–2488, 2017. doi:10.1049/iet-gtd.2016.0968.

- [11] J. E. Siegel, S. R. Pratt, Y. Sun, and S. E. Sarma, "Real-Time Deep Neural Networks for Internet-Enabled Arc-Fault Detection," *Engineering Applications of Artificial Intelligence*, vol. 74, pp. 35–42, 2018.
doi:10.1016/j.engappai.2018.05.009.
- [12] L. K. Baxter, *Capacitive Sensors: Design and Applications*. Piscataway, NJ, USA: IEEE Press, 1997.
- [13] M. N. O. Sadiku, *Fundamentals of Electric Circuits*, 6th ed. New York, NY, USA: McGraw-Hill, 2017.
- [14] P. Horowitz and W. Hill, *The Art of Electronics*, 3rd ed. Cambridge, UK: Cambridge University Press, 2015.
- [15] Omron Corporation, *G3M Series Solid State Relay Datasheet*, Omron Electronic Components LLC.
- [16] Microchip Technology Inc., *ATmega328P Datasheet*, Microchip Technology Inc., 2020.
- [17] N. Mohan, T. M. Undeland, and W. P. Robbins, *Power Electronics: Converters, Applications, and Design*, 3rd ed. Hoboken, NJ, USA: Wiley, 2003.

APPENDICES

Supplementary project materials and implementation resources

A. Experiment Videos

Short demonstration videos documenting the prototype tests and observed system behaviour.

B. Arduino Code

The complete Arduino/ATmega328P firmware used for signal sampling, adaptive thresholding, decision logic, and relay control.

C. Components Used

A concise summary of the electronic components and modules used to build and test the prototype.

More details and project resources

Project resource page:
<https://www.motarig.com/academic/en>

Scan the QR code to access the full supplementary materials.
The page provides the videos, source code, and component information in one place.

